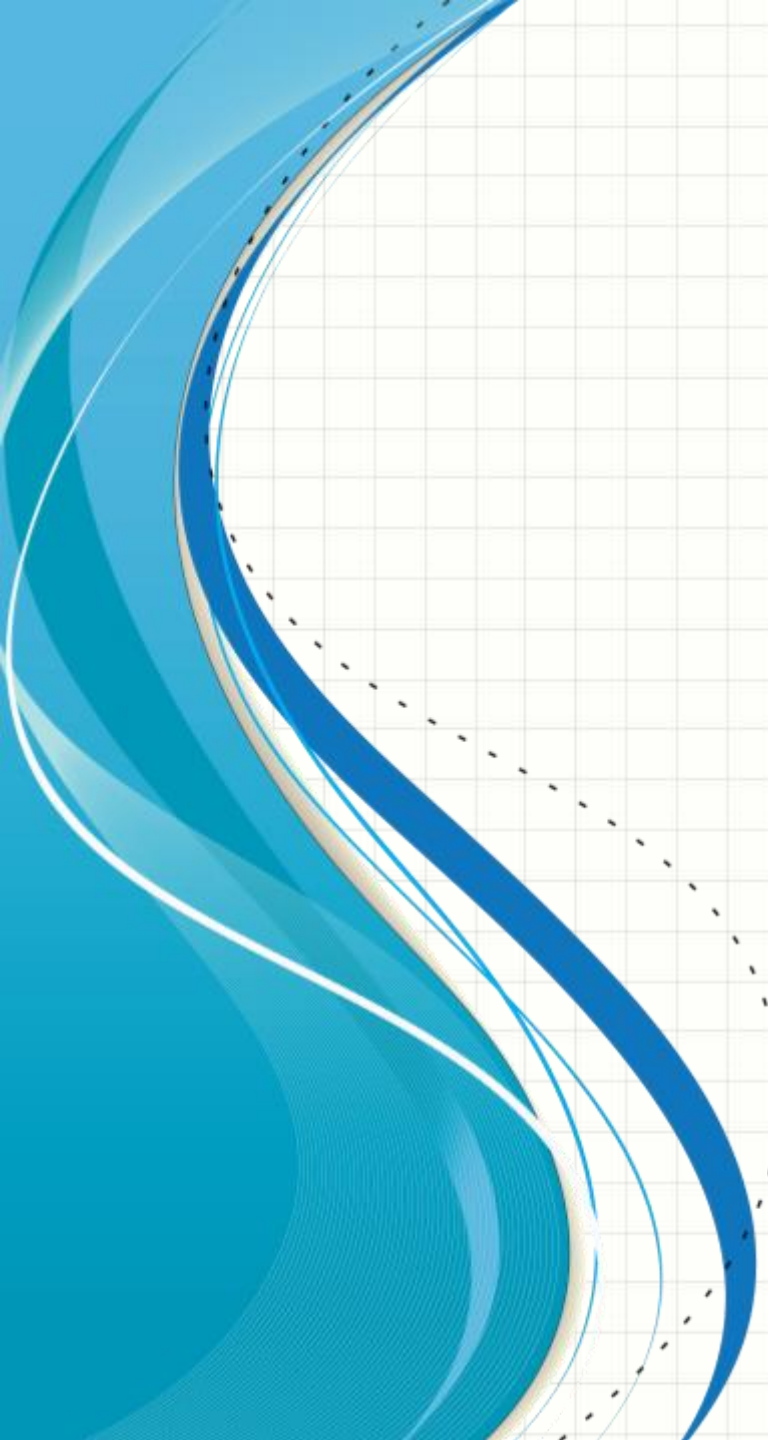




CSE-251

ELECTRONIC CIRCUITS

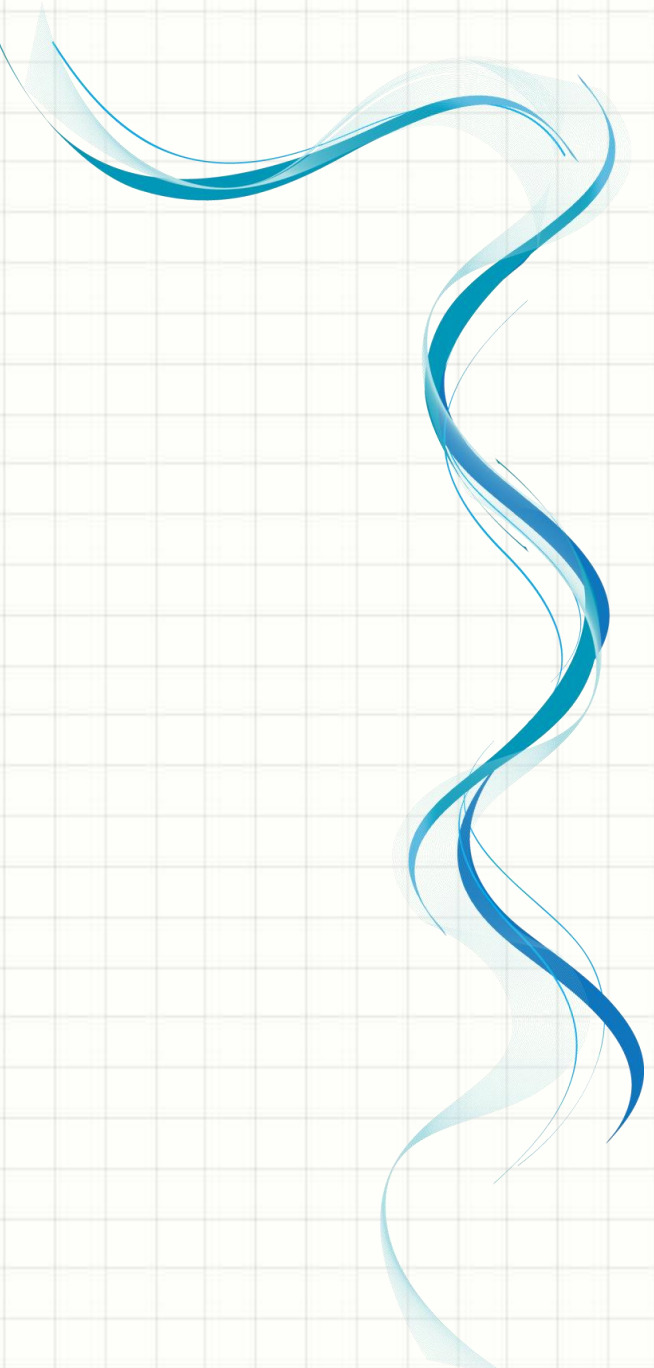
LECTURE-06

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Today's Overview:

- ☐ *Static and Dynamic Resistances*
- ☐ *Ideal Diode and Constant Voltage Drop Models*
- ☐ *Load Line Analysis*



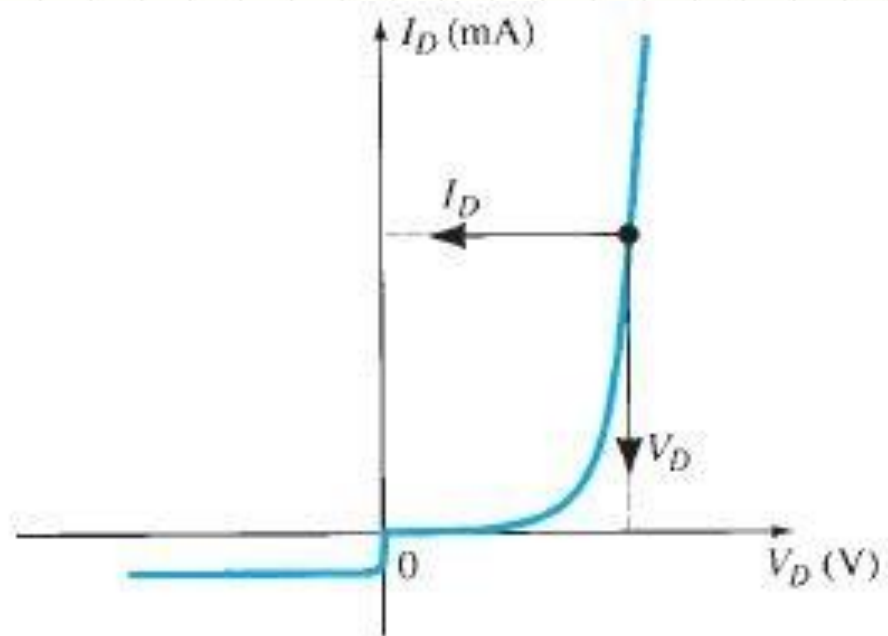
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Static and Dynamic Resistances

DC or Static Resistance:

The opposition that is felt by the electric current when an external DC voltage is applied to the p-n junction is called the Static resistance.

$$R_D = \frac{V_D}{I_D}$$

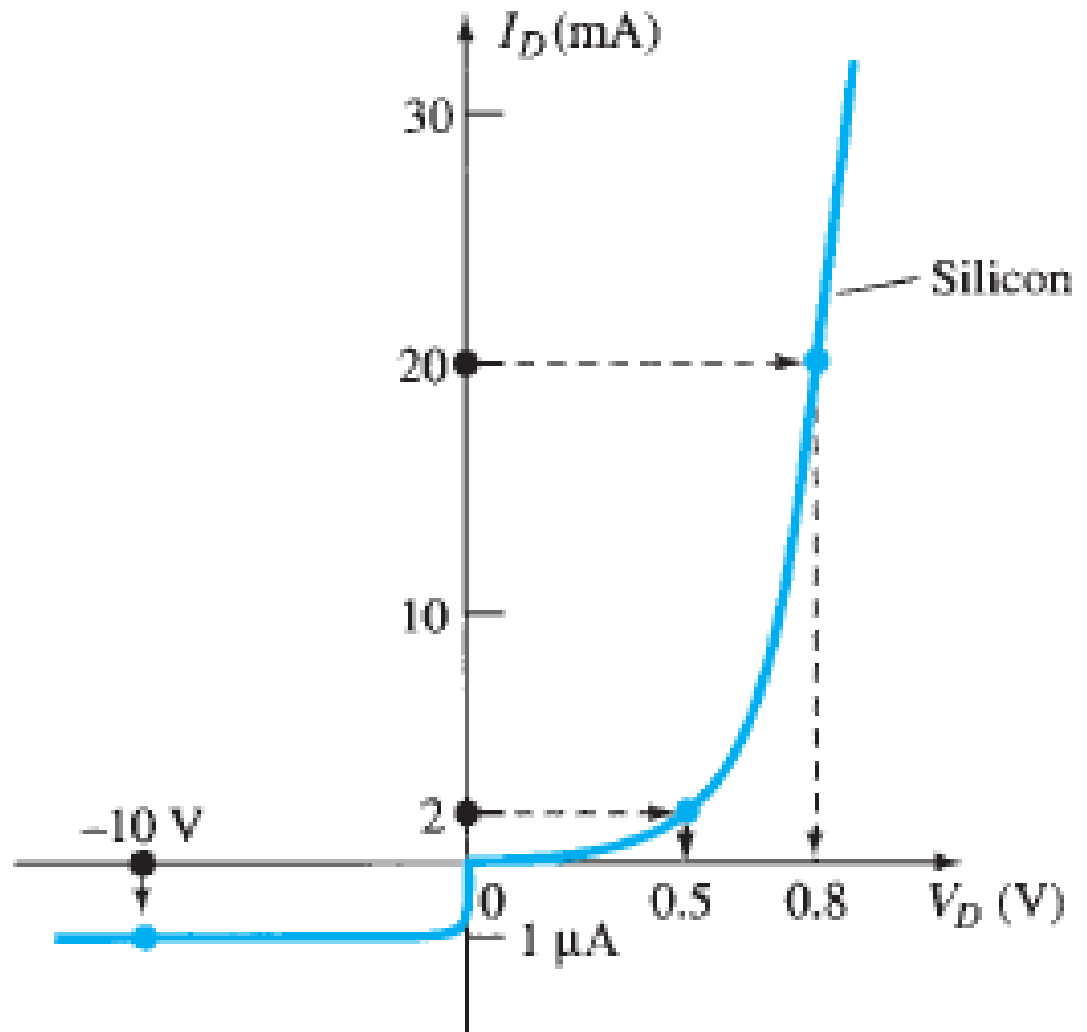


Determining the dc resistance of a diode at a particular operating point.

Problem:

□ Determine the Dc resistance levels for the diode at-

- a) $I_D = 2\text{mA}$
- b) $I_D = 20\text{mA}$
- c) $V_D = -20\text{V}$



Problem:

Solution:

a. At $I_D = 2 \text{ mA}$, $V_D = 0.5 \text{ V}$ (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.5 \text{ V}}{2 \text{ mA}} = 250 \, \Omega$$

b. At $I_D = 20 \text{ mA}$, $V_D = 0.8 \text{ V}$ (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{0.8 \text{ V}}{20 \text{ mA}} = 40 \, \Omega$$

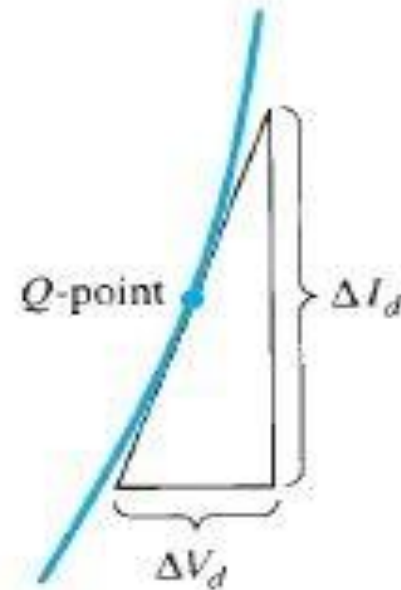
c. At $V_D = -10 \text{ V}$, $I_D = -I_s = -1 \, \mu\text{A}$ (from the curve) and

$$R_D = \frac{V_D}{I_D} = \frac{10 \text{ V}}{1 \, \mu\text{A}} = 10 \text{ M}\Omega$$

AC or Dynamic Resistances:

The opposition that is felt by the electric current when an external AC voltage is applied to the p-n junction is called the AC or Dynamic resistance.

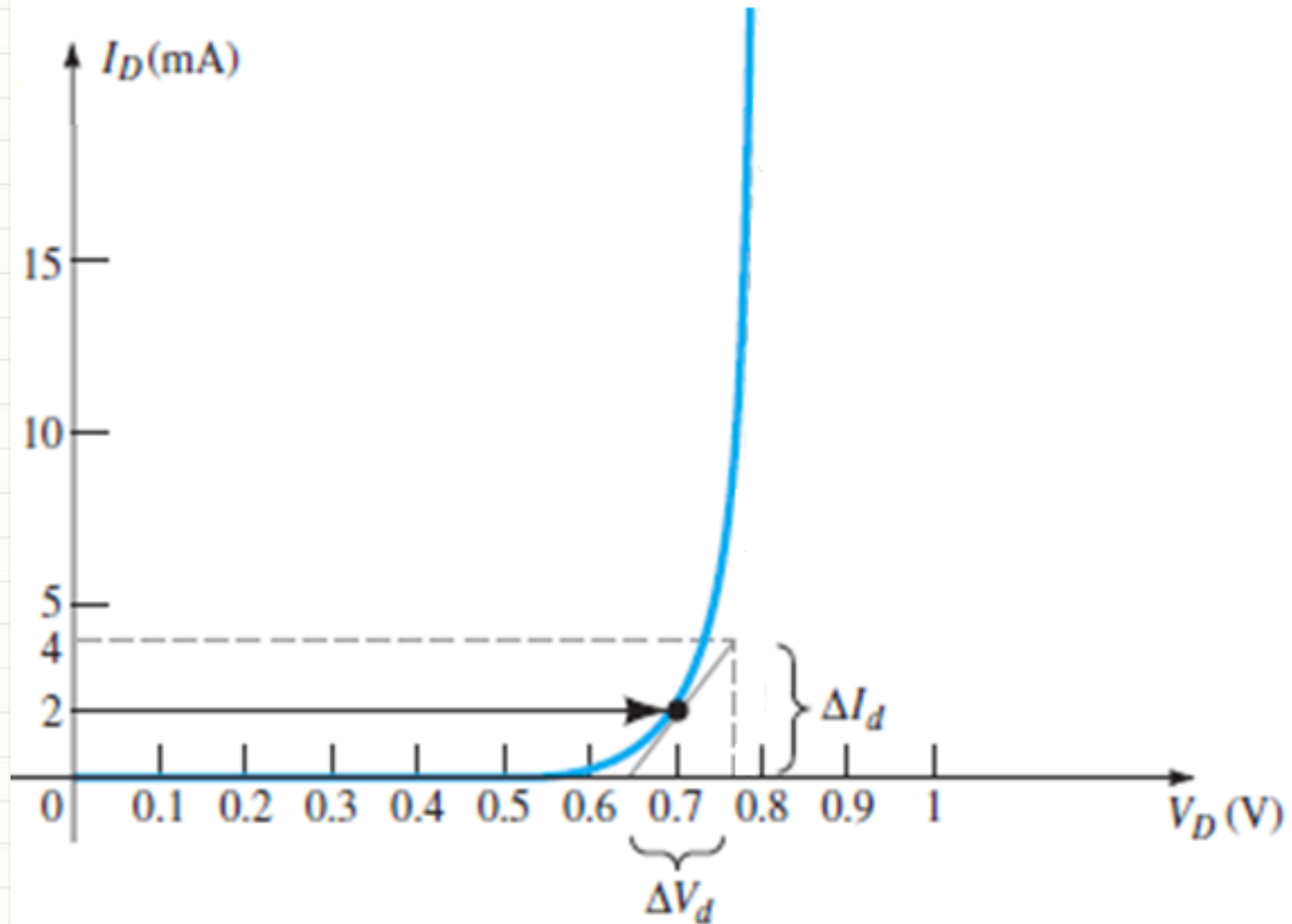
$$r_d = \frac{\Delta V_d}{\Delta I_d}$$



Determining the ac resistance at a Q-point.

Problem:

- Determine the ac resistance at $I_D = 2 \text{ mA}$.



Problem:

Solution:

$$\Delta I_d = 4 \text{ mA} - 0 \text{ mA} = 4 \text{ mA}$$

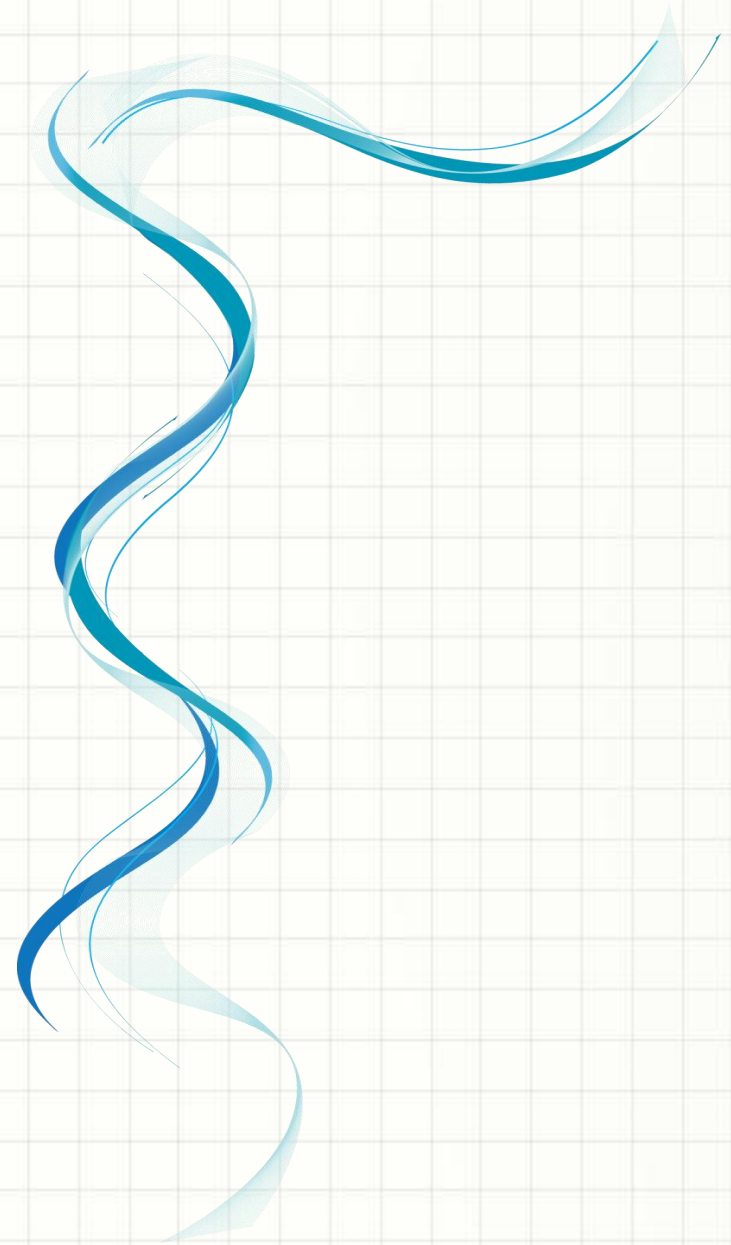
and

$$\Delta V_d = 0.76 \text{ V} - 0.65 \text{ V} = 0.11 \text{ V}$$

and the ac resistance is

$$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.11 \text{ V}}{4 \text{ mA}} = 27.5 \Omega$$

Ideal Diode and Constant Voltage Drop Models



Ideal Diode Model:

In this model, we assume that there is no voltage drop across the depletion layer of the diode. This model also ignores the diode's dynamic resistance. As a result, the diode behaves like a complete **short circuit** in the forward bias region.

According to the ideal diode model, we can consider a forward-biased ideal diode as a **wire** in the circuit. Then, the current will flow through without any resistance in the circuit according to Ohm's law.

Similarly, the current flowing through the ideal diode in the reverse bias region is assumed to be zero. As a result, the ideal diode behaves like an **open circuit** in the reverse bias region. If an ideal diode is in the reverse bias region in a circuit, we assume that there is a **break** in that part of the circuit.

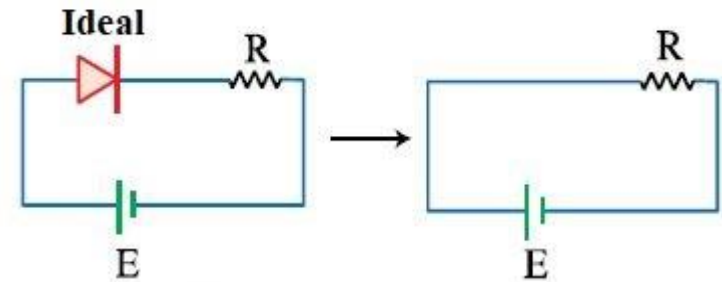


Fig : Forward bias in Ideal Diode

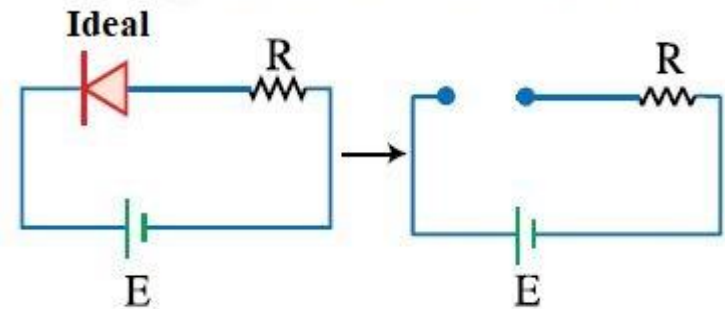


Fig : Reverse bias in Ideal Diode

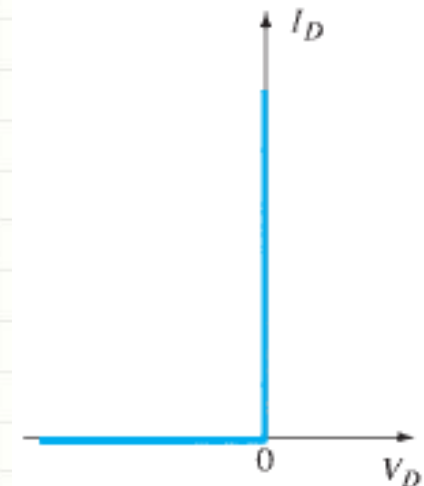
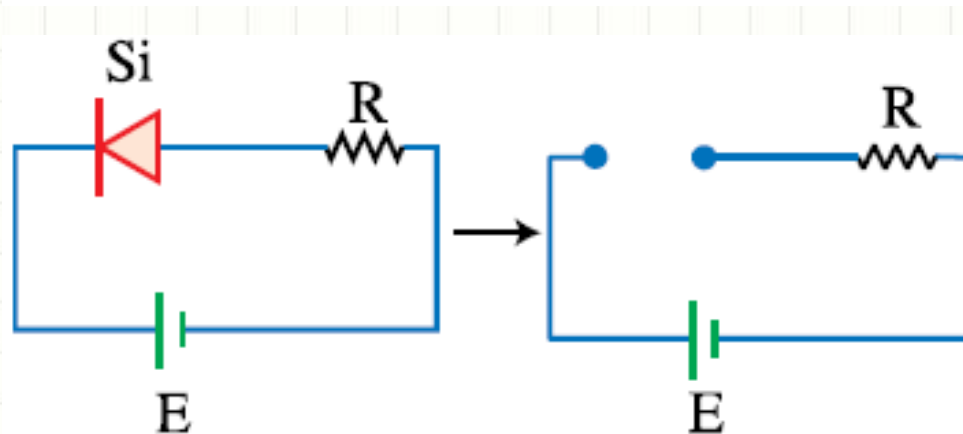
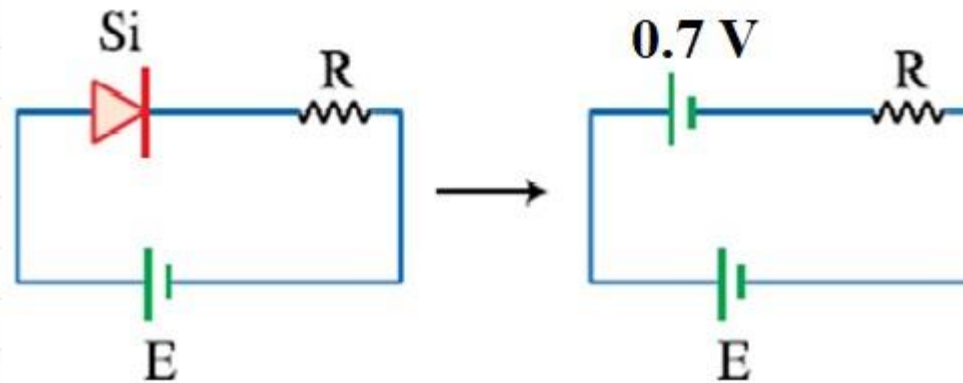


Fig : Characteristics Curve of I-V flow in Ideal Diode

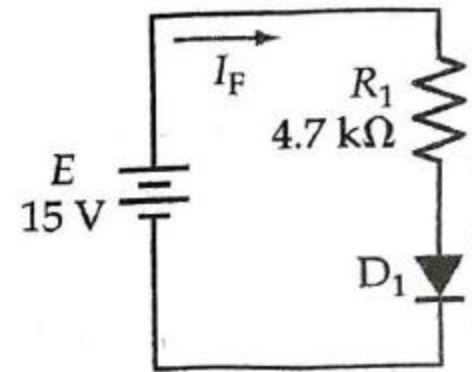
Constant Voltage Drop Model:

In this model, we will not ignore the barrier of the depletion layer of the diode. If forward bias is applied to the diode in the circuit, the diode uses some voltage to overcome the barrier of its depleted layer, and then the current starts to flow. In the case of the constant voltage drop model, the diode will always have the same voltage drop, regardless of the current in the circuit.



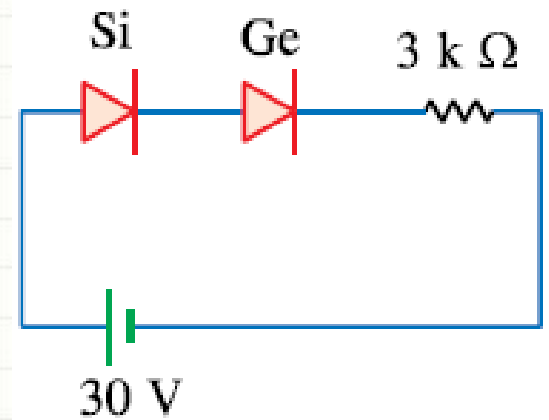
Problem:

Calculate the voltage across and current through the resistor. [Consider the diode made of Si]



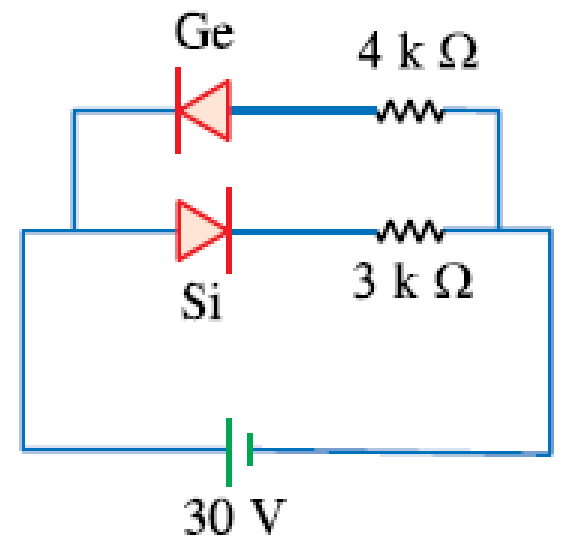
Problem:

Calculate I_F for the circuit.



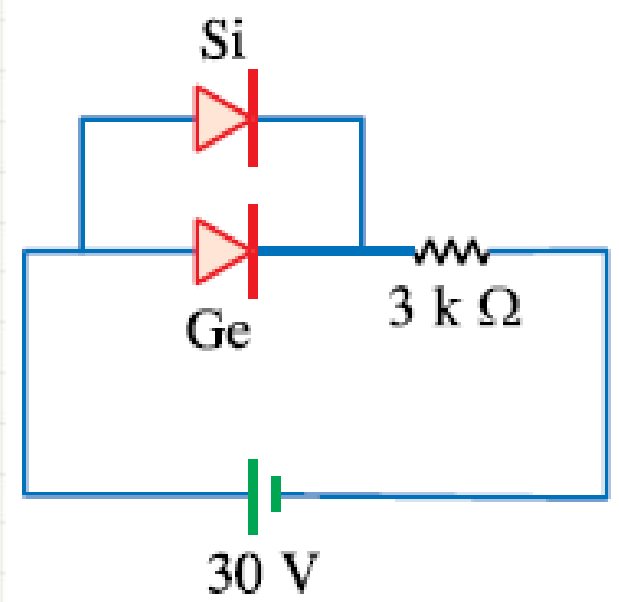
Problem:

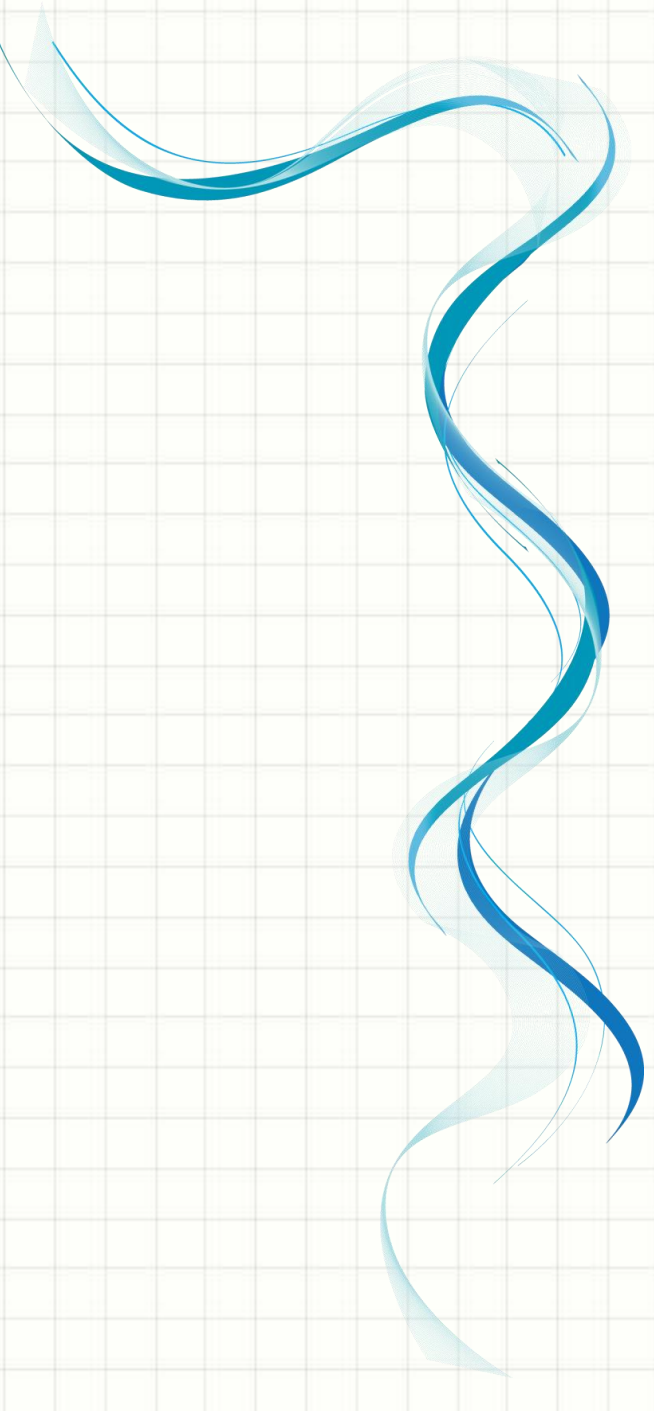
Calculate the current circulating the circuit.



Problem:

Calculate the current circulating the circuit.



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Load Line Analysis

Load Line:

A load line is a line drawn on the current–voltage characteristic graph for a nonlinear device like a diode or transistor.

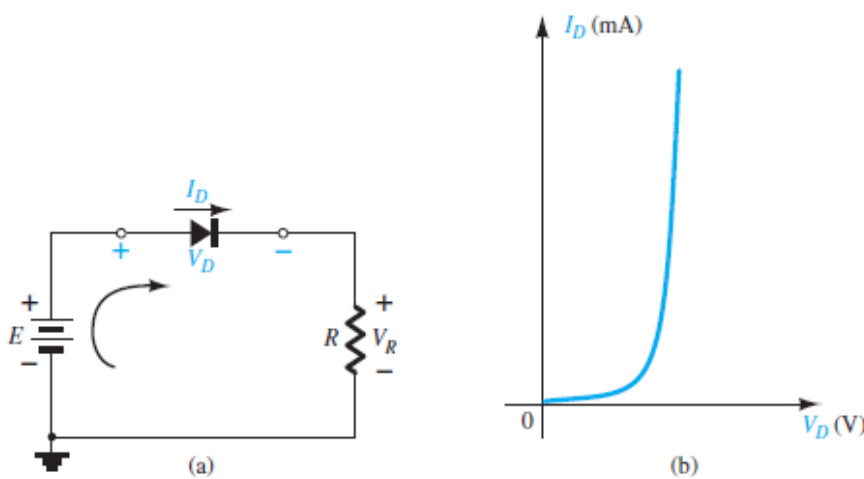


FIG. 2.1

Series diode configuration: (a) circuit; (b) characteristics.

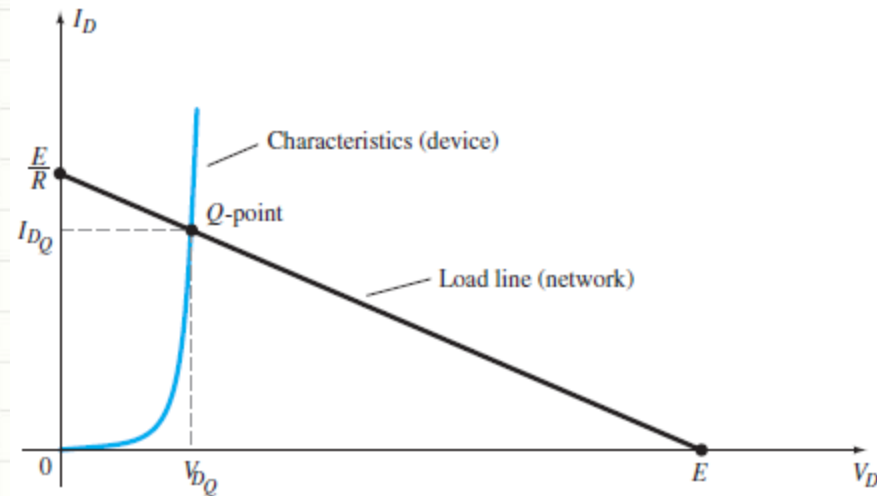


FIG. 2.2

Drawing the load line and finding the point of operation.

Load Line Analysis:

The intersections of the load line on the characteristics of Fig. 2.2 can be determined by first applying Kirchhoff's voltage law:

$$E = V_D + I_D R$$

The intersections of the load line on the characteristics can easily be determined if one simply employs the fact that anywhere on the horizontal axis $I_D = 0$ A and anywhere on the vertical axis $V_D = 0$ V.

If we set $V_D = 0$ V in the eq. above and solve for I_D , we have the magnitude of I_D on the vertical axis. Therefore, $E = V_D + I_D R = 0 + I_D R$. So,

$$I_D = \frac{E}{R} \Big|_{V_D=0 \text{ V}}$$

If we set $I_D = 0$ A in the eq. and solve for V_D , we have the magnitude of V_D on the horizontal axis. Therefore, $E = V_D + I_D R = V_D + 0.R$. So,

$$V_D = E \Big|_{I_D=0 \text{ A}}$$

The point of operation is usually called the *quiescent point* (abbreviated “**Q - point**”) to reflect its “still, unmoving” qualities as defined by a dc network.

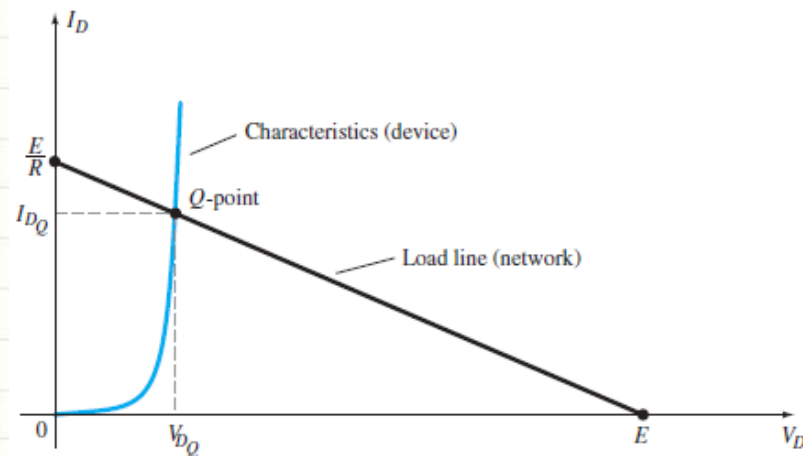


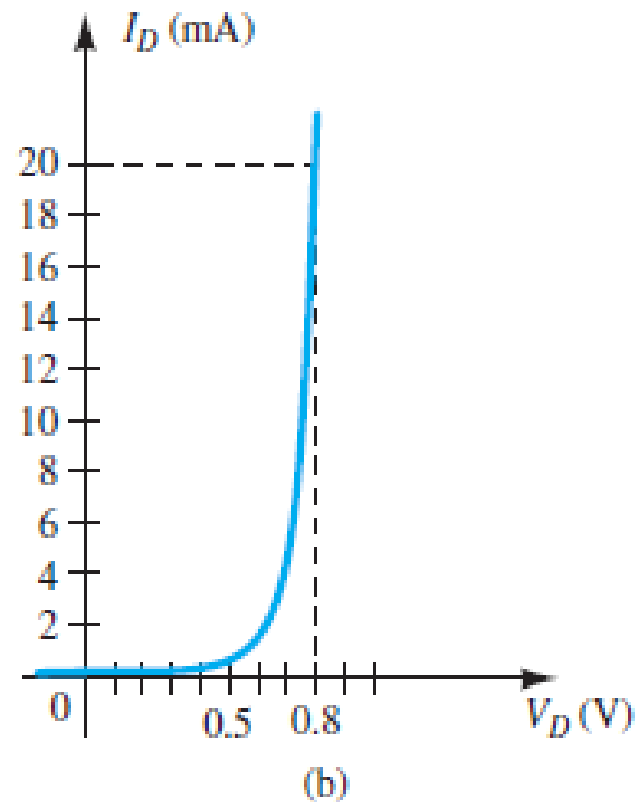
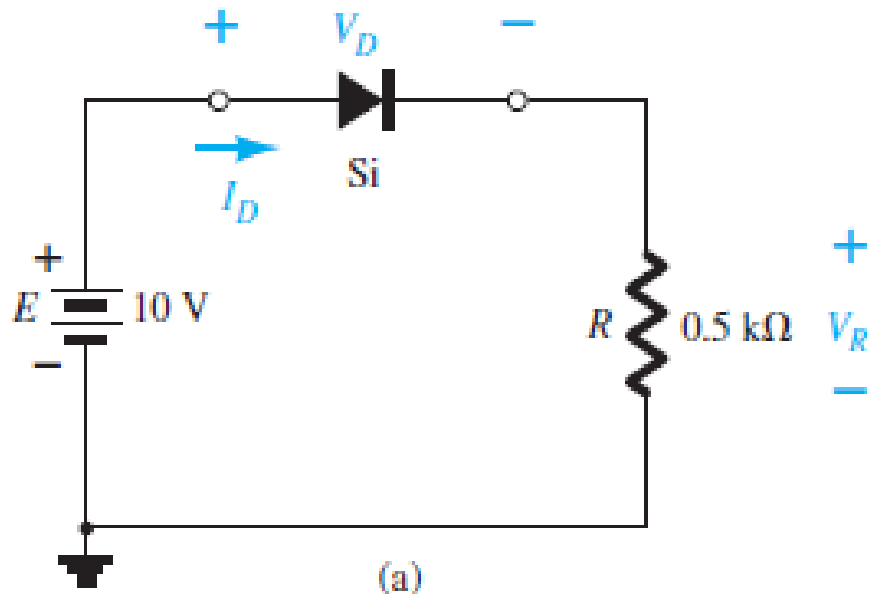
FIG. 2.2

Drawing the load line and finding the point of operation.

Problem:

For the series diode configuration, determine:

- V_{DQ} and I_{DQ} .
- V_R .



Problem:

Solution:

a. Eq. (2.2): $I_D = \frac{E}{R} \Big|_{V_D=0\text{ V}} = \frac{10\text{ V}}{0.5\text{ k}\Omega} = 20\text{ mA}$

Eq. (2.3): $V_D = E \Big|_{I_D=0\text{ A}} = 10\text{ V}$

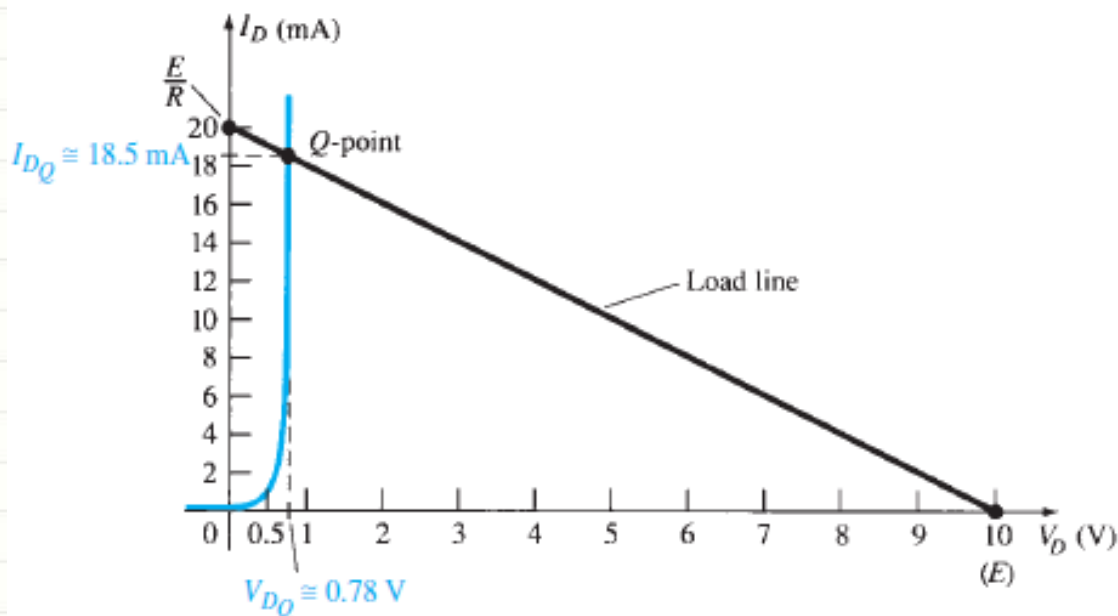
The resulting load line appears in Fig. 2.4. The intersection between the load line and the characteristic curve defines the Q -point as

$$V_{DQ} \cong 0.78\text{ V}$$

$$I_{DQ} \cong 18.5\text{ mA}$$

The level of V_D is certainly an estimate, and the accuracy of I_D is limited by the chosen scale. A higher degree of accuracy would require a plot that would be much larger and perhaps unwieldy.

b. $V_R = E - V_D = 10\text{ V} - 0.78\text{ V} = 9.22\text{ V}$





QUESTIONS?

SEE YOU IN THE NEXT CLASS

